

Confidence limits for difference of means

If $\bar{x}_1$ and $\bar{x}_2$ are the sample means based on large and independent random samples of sizes n_1 & n_2 from two infinite populations with means μ_1 & μ_2 and standard deviations are σ_1 and σ_2 respectively, then

$$E(\bar{x}_1 - \bar{x}_2) = E(\bar{x}_1) - E(\bar{x}_2) = \mu_1 - \mu_2$$

$$\begin{aligned} \text{Var}(\bar{x}_1 - \bar{x}_2) &= \text{Var}(\bar{x}_1) + \text{Var}(\bar{x}_2) \\ &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \end{aligned}$$

Confidence interval for difference in means is given by

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \cdot SE(\bar{x}_1 - \bar{x}_2)$$

$$\text{i.e., } (\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

If σ_1^2 and σ_2^2 are unknown, then their estimate provided by the corresponding sample variances s_1^2 and s_2^2 are used. Then the confidence limits become,

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Q. Find 95% confidence interval for the difference between means for the data:

$$\bar{x}_1 = 88.42, \quad s_1 = 5.62, \quad n_1 = 50$$

$$\bar{x}_2 = 80.61, \quad s_2 = 4.83, \quad n_2 = 50$$

Solution

$$\text{Given } \bar{x}_1 = 88.42, \quad s_1 = 5.62, \quad n_1 = 50$$

$$\bar{x}_2 = 80.61, \quad s_2 = 4.83, \quad n_2 = 50$$

$$100(1-\alpha)\% = 95\%$$

$$\alpha = 0.05 \quad \alpha/2 = 0.025 \therefore Z_{\alpha/2} = 1.96$$

Confidence Interval on difference of mean is given by

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\text{ie } (88.42 - 80.61) \pm 1.96 \cdot \sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}$$

$$\text{ie, } 7.81 \pm 1.96 \cdot \sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}$$

$$\text{ie, } 7.81 \pm 2.05$$

$$\text{ie } 7.81 - 2.05 \leq (\mu_1 - \mu_2) \leq 7.81 + 2.05$$

$$\text{ie } \underline{5.76 \leq (\mu_1 - \mu_2) \leq 9.83}$$

Q. A sample survey of 40 small town coffee shops and 49 big city coffee shops, and find that the mean price of a large cup of coffee is Rs. 3.75 and in the big cities it is Rs. 4.5. The population standard deviation for small town is 1.2 & in big cities population standard deviation is 0.98. Construct 99% confidence interval for the difference of their two means & draw conclusions from it.

Given $n_1 = 40$ $\bar{x}_1 = 3.75$ $\sigma_1 = 1.2$
 $n_2 = 49$ $\bar{x}_2 = 4.5$ $\sigma_2 = 0.98$

$100(1-\alpha)\% = 99\%$
 $\alpha = 0.01$ $\therefore \alpha/2 = 0.005$

$\therefore Z_{\alpha/2} = 2.58$

C.I on difference of mean is given by

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

as population sd is given we use σ instead of s

$$(3.75 - 4.5) \pm 2.58 \cdot \sqrt{\frac{1.2^2}{40} + \frac{0.98^2}{49}}$$

$$-0.75 \pm 2.58 \cdot \sqrt{0.036 + 0.0196}$$

$$-0.75 \pm 0.61$$

$$\text{i.e. } -0.75 - 0.61 \leq (\mu_1 - \mu_2) \leq -0.75 + 0.61$$

$$-1.36 \leq (\mu_1 - \mu_2) \leq -0.14$$

Find 95% C.I

$$-1.21 \leq (\mu_1 - \mu_2) \leq -0.29$$